3. Survey Report on Emerging Trends in Low-Code Application Development

Aim:

To study and analyse the emerging trends in low-code application development and understand their impact on modern software development.

Theory:

Low-code application development is a modern software development approach that enables users to create applications using visual interfaces, drag-and-drop components, and minimal hand-written code. It accelerates application development, reduces costs, and enables both professional developers and citizen developers to build business applications efficiently.

With the increasing demand for digital transformation, low-code platforms are continuously evolving. Some of the major emerging trends in low-code application development include:

- **Artificial Intelligence (AI) Integration:** AI-powered features such as code suggestions, workflow automation, chatbots, and predictive analytics improve development efficiency.
- **Citizen Development:** Non-technical users can develop business applications with minimal programming knowledge.
- **Cloud-Native Development:** Most low-code platforms are cloud-based, enabling easy deployment, scalability, and remote collaboration.
- **Business Process Automation:** Integration with workflow automation helps organizations streamline repetitive business tasks.
- **Cross-Platform Application Development:** Applications can be developed once and deployed across web and mobile platforms.
- **API and Third-Party Integration:** Low-code platforms support seamless integration with external databases, cloud services, and enterprise applications.
- **Enhanced Security and Governance:** Modern platforms provide role-based access control, data encryption, audit logs, and compliance features.
- **DevOps and Continuous Deployment:** Integration with CI/CD pipelines enables faster testing, deployment, and application updates.
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These trends are making low-code platforms an important technology for rapid application development and enterprise digital transformation.

Procedure:

1. Study the concept of low-code application development.
2. Identify popular low-code platforms and their key features.
3. Collect information on the latest trends in low-code development from reliable sources such as research papers, industry reports, and official documentation.
4. Analyze trends such as AI integration, citizen development, cloud computing, automation, API integration, and security.
5. Compare the benefits and challenges associated with each trend.
6. Record the observations and summarize the findings.
7. Prepare the survey report based on the collected information.

Result:

The survey revealed that low-code application development is rapidly evolving with trends such as AI-assisted development, cloud-native platforms, citizen development, business process automation, API integration, and enhanced security. These advancements are improving development speed, reducing costs, and enabling organizations to build scalable applications more efficiently while supporting digital transformation initiatives.